

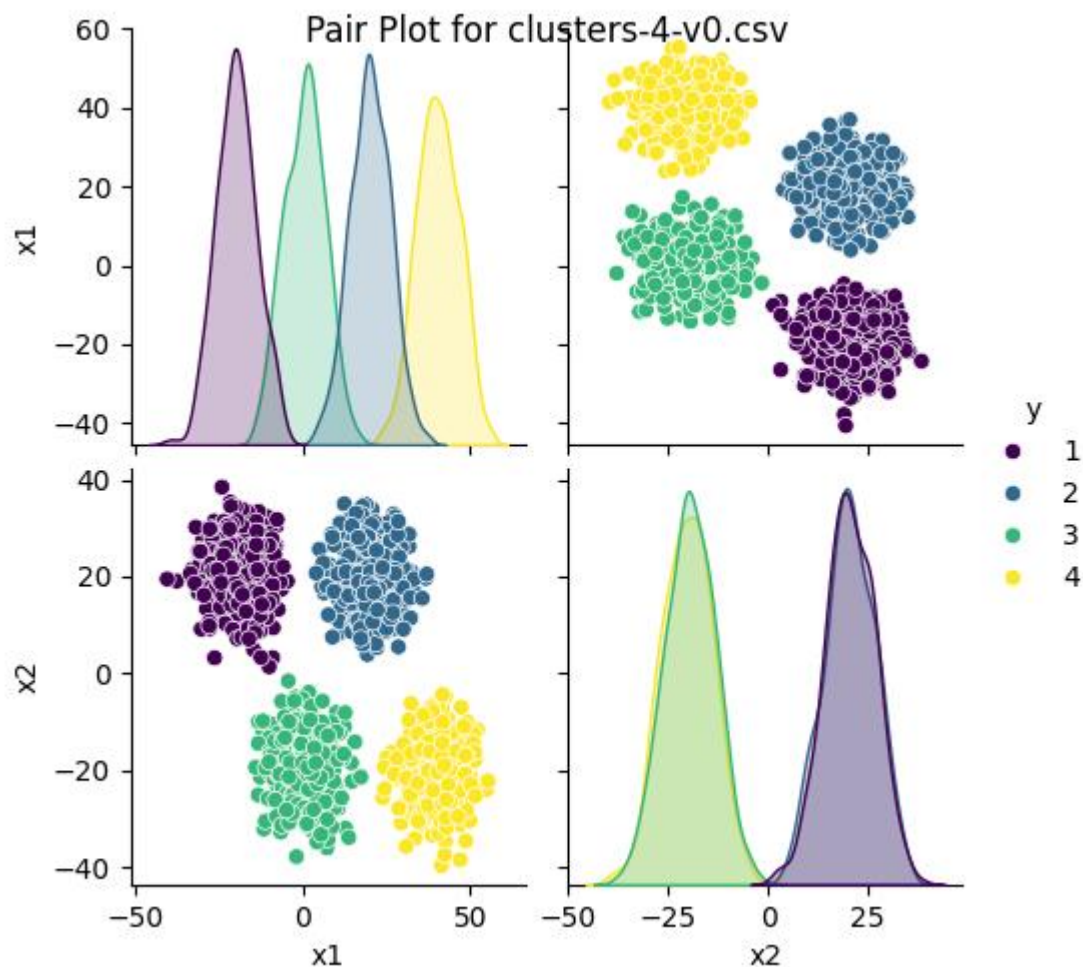
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Part – 1:

1.



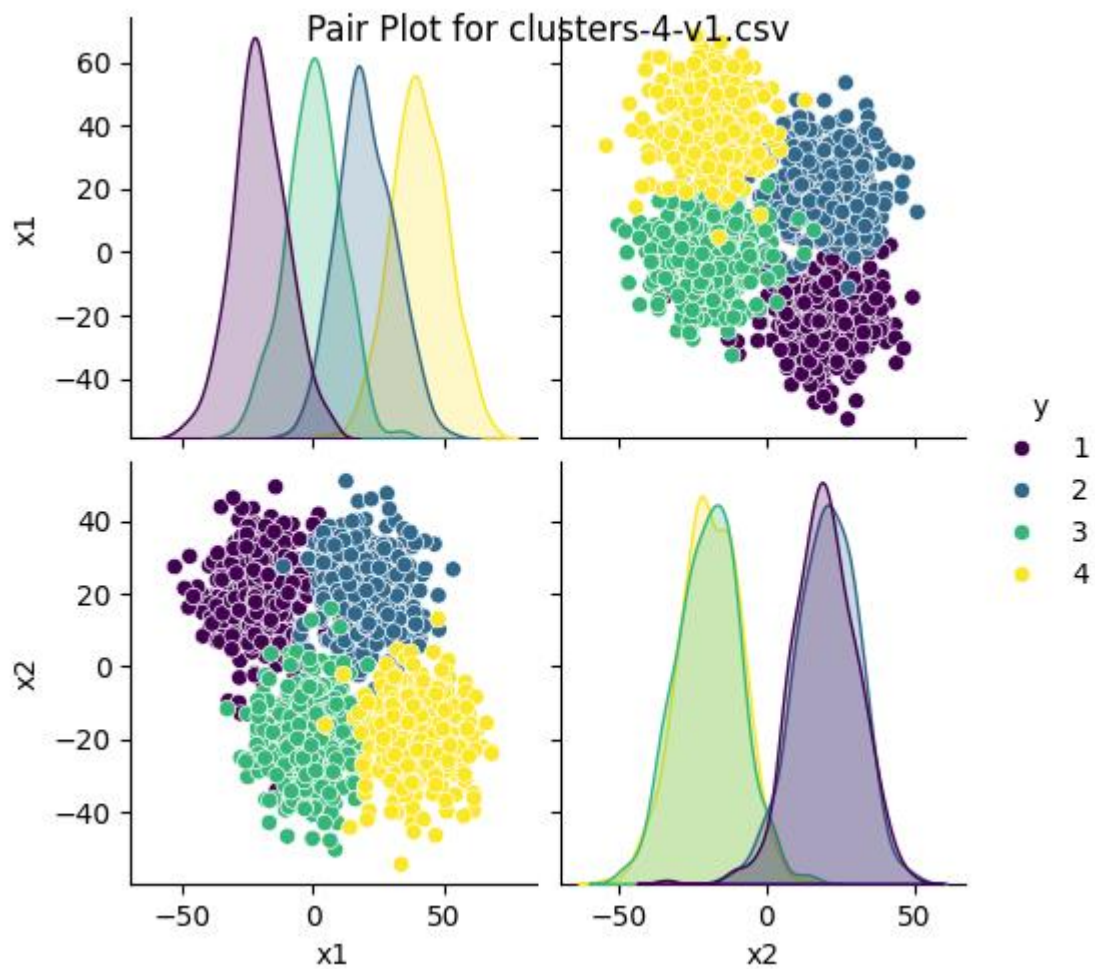
This pair plot for the dataset of clusters-4-v0

Plots on the diagonal show the relationship between variables x1 and x2 using the scatter plot. Other plots show the distribution for each variable.

From the plot, we can see that the four clusters are well separated in both x1 and x2.

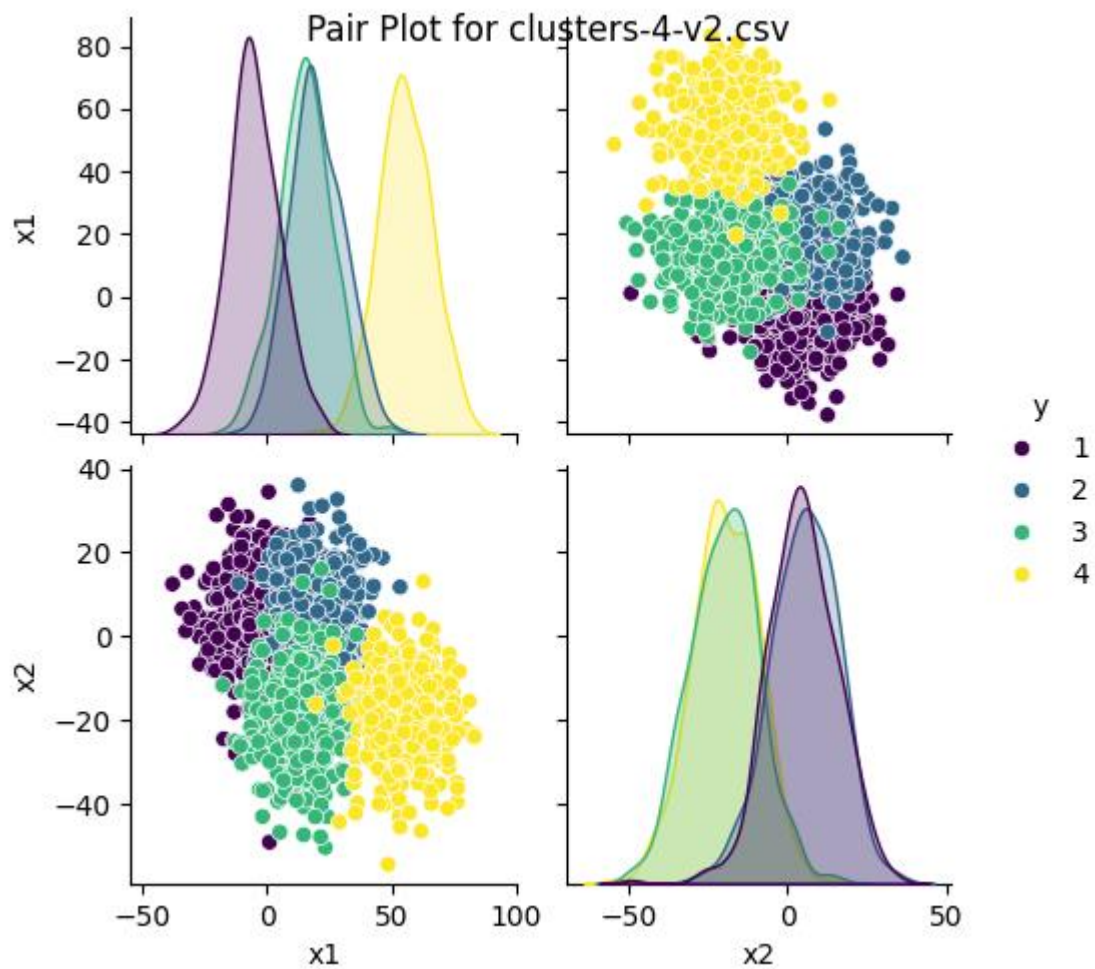
The Distribution plot for x1 shows that Clusters are well separated for x1, and x1 alone can do a good job of classifying four clusters.

2.



For the second dataset, we can see that clusters are much harder to classify, but we can still see a clear distinction between each cluster. x_1 is can still be used alone for classification, but it won't as accurate as for the last dataset. Thus, both x_1 and x_2 are required for a good classification.

3.



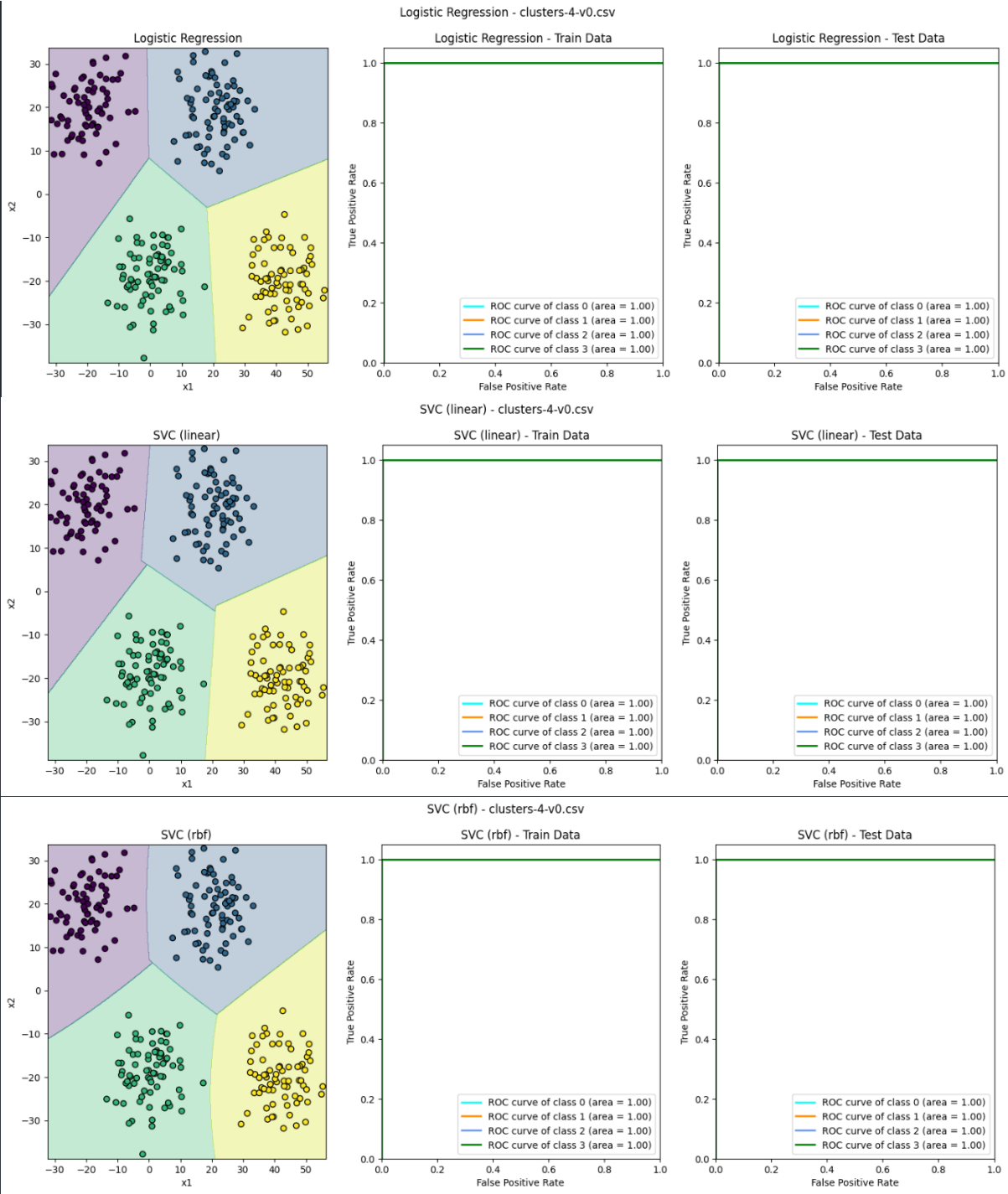
This dataset is much more difficult to classify than the previous two, with clusters overlapping significantly. Neither x_1 nor x_2 alone can be used for a good classification.

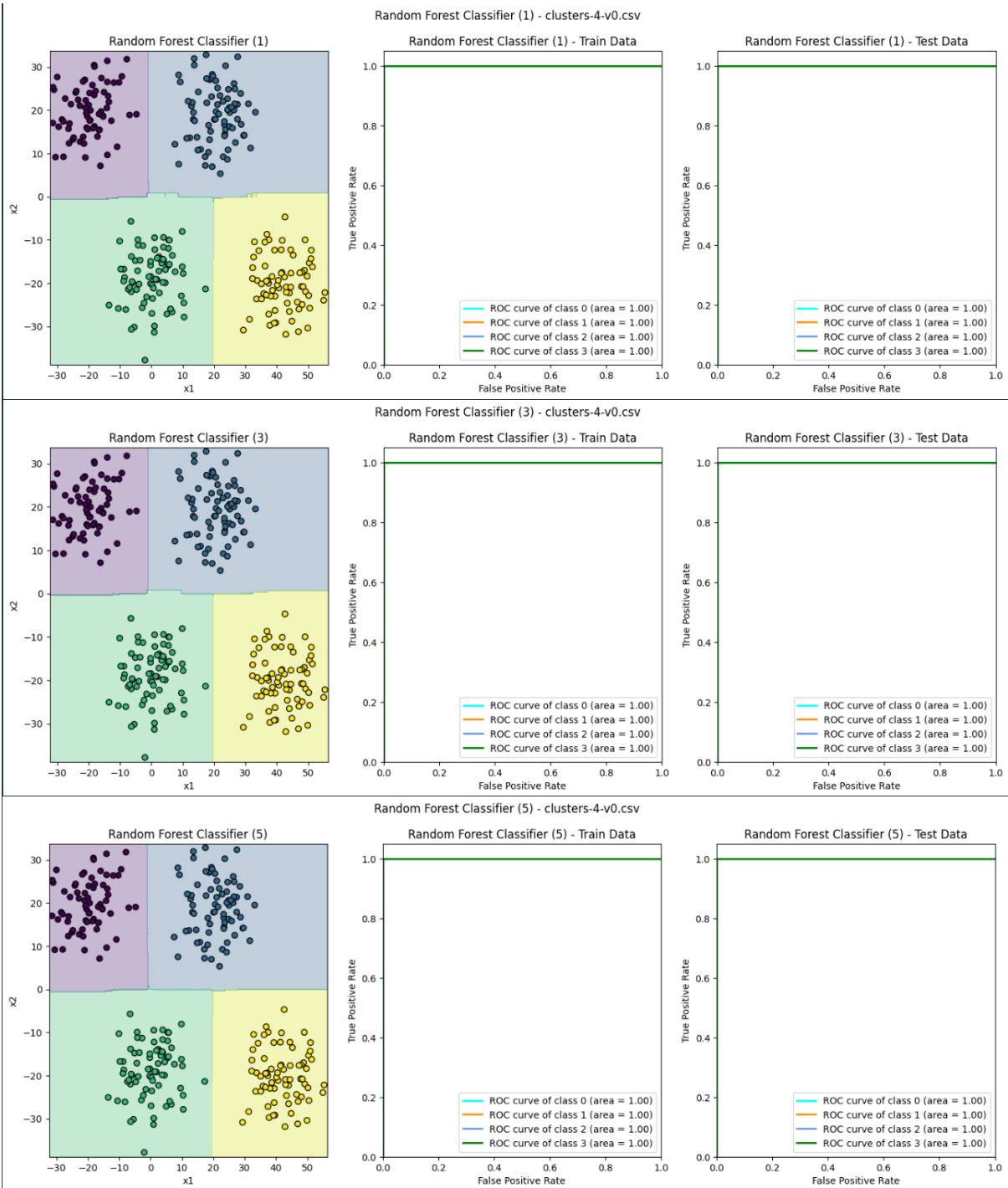
By Observing all three datasets, we can comfortably say that Logistic Regression will perform best on clusters-4-v0.csv

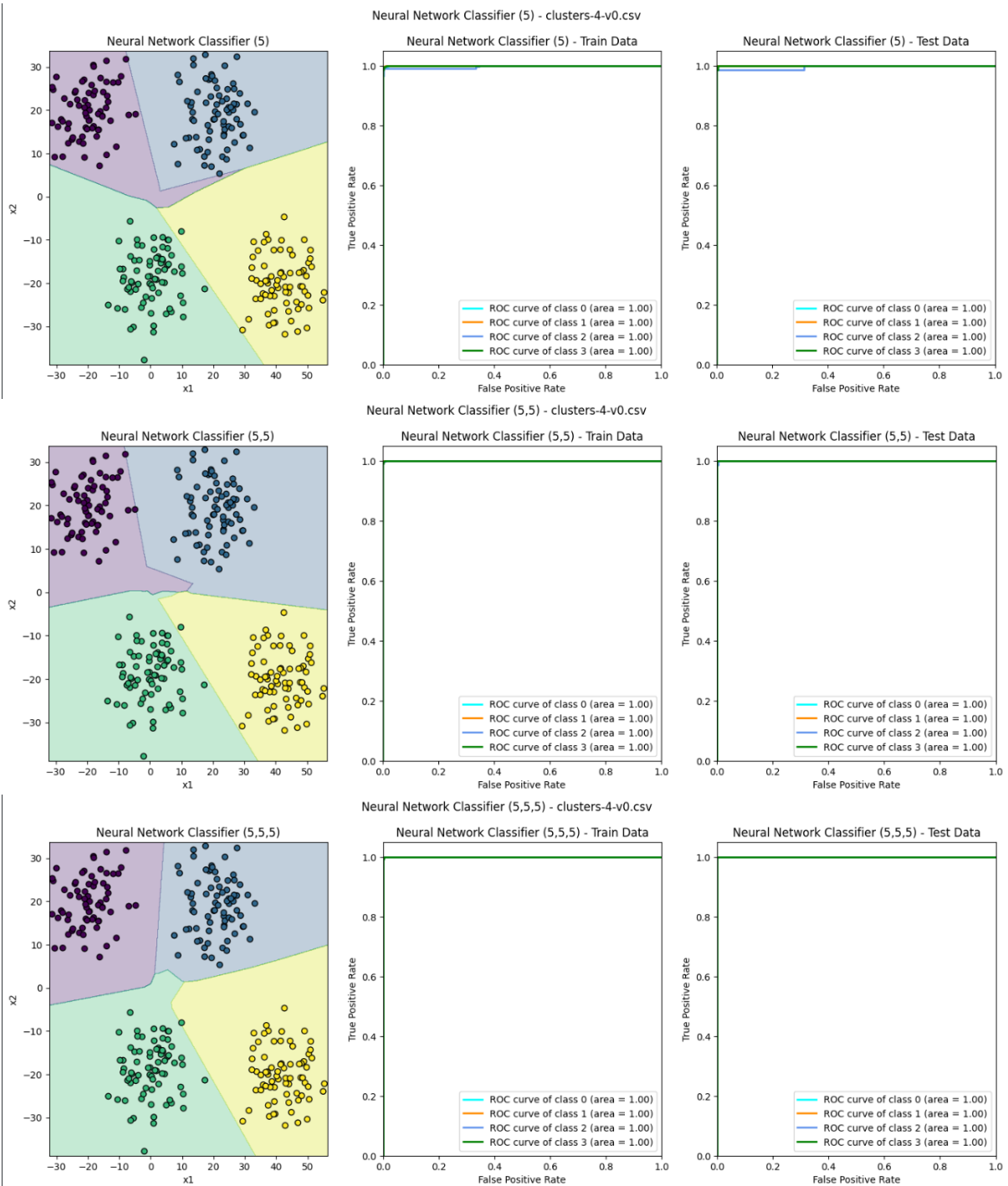
Part – 3:

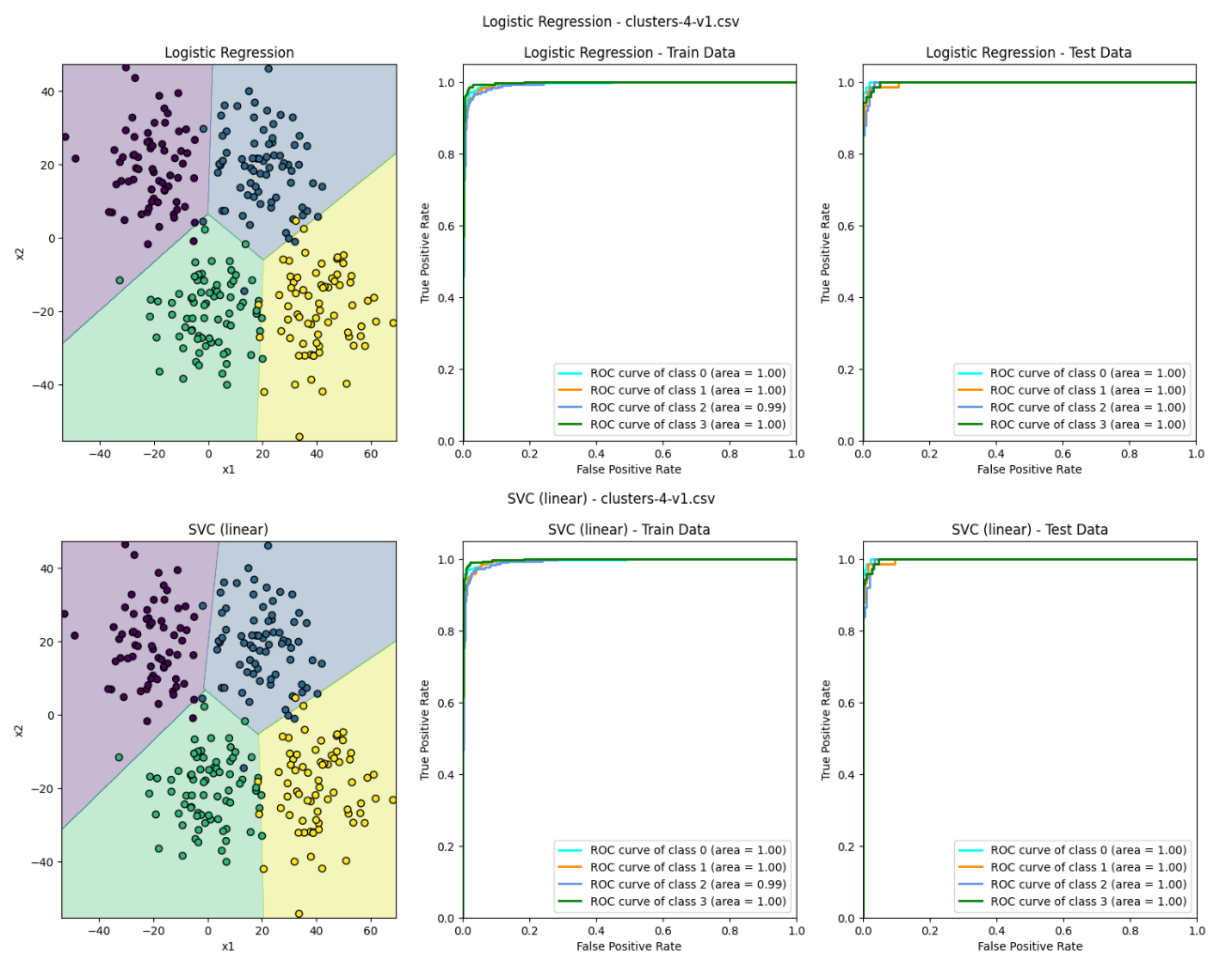
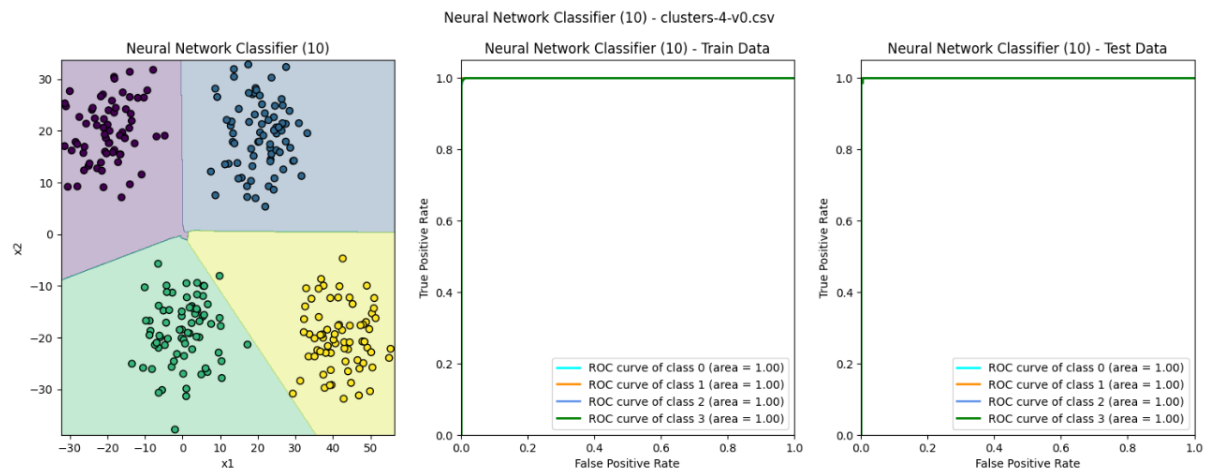
file	model	dataset	accuracy	precision_1	precision_2	precision_3	precision_4	precision_avg	recall_1	recall_2	recall_3	recall_4	recall_avg	f1_1	f1_2	f1_3	f1_4	f1_avg	auc_1	auc_2	auc_3	auc_4	auc_avg
clusters-4-v0.csv	Logistic Regression	train	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	Logistic Regression	test	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	SVC (linear)	train	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	SVC (linear)	test	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	SVC (rbf)	train	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	SVC (rbf)	test	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	Random Forest Classifier (1)	train	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	Random Forest Classifier (1)	test	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	Random Forest Classifier (3)	train	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	Random Forest Classifier (3)	test	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	Random Forest Classifier (5)	train	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	Random Forest Classifier (5)	test	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v0.csv	Neural Network Classifier (5)	train	0.992188	0.99657534	1	0.99293286	0.97959184	0.99227501	0.996575	0.992982	0.982517	0.99654	0.992153768	0.996575	0.996479	0.987698	0.987993	0.992186	0.999928	0.999964	0.996374	0.999908	0.999043
clusters-4-v0.csv	Neural Network Classifier (5)	test	0.996528	1	1	1	0.98611111	0.99652778	1	1	0.986486	1	0.996621622	1	1	0.993197	0.993007	0.996551	1	1	0.995769	0.999935	0.998926
clusters-4-v0.csv	Neural Network Classifier (5,5)	train	0.99566	1	1	0.99646643	0.98630137	0.99569195	1	1	0.986014	0.99654	0.995638445	1	1	0.991213	0.991394	0.995652	1	1	0.999931	0.999948	0.99997
clusters-4-v0.csv	Neural Network Classifier (5,5)	test	0.996528	1	1	1	0.98611111	0.99652778	1	1	0.986486	1	0.996621622	1	1	0.993197	0.993007	0.996551	1	1	0.999937	1	0.999984
clusters-4-v0.csv	Neural Network Classifier (5,5,5)	train	0.996528	1	1	0.99647887	0.98969072	0.996542399	1	1	0.98951	0.99654	0.99551257	1	1	0.992982	0.993103	0.996521	1	1	0.99998	0.99998	0.99999
clusters-4-v0.csv	Neural Network Classifier (5,5,5)	test	0.996528	1	1	1	0.98611111	0.99652778	1	1	0.986486	1	0.996621622	1	1	0.993197	0.993007	0.996551	1	1	1	1	1
clusters-4-v0.csv	Neural Network Classifier (10)	train	0.994792	1	1	0.9964539	0.98293515	0.994847264	1	1	0.982517	0.99654	0.994764319	1	1	0.989437	0.989691	0.994782	1	1	0.999911	0.99994	0.999963
clusters-4-v0.csv	Neural Network Classifier (10)	test	0.996528	1	1	1	0.98611111	0.99652778	1	1	0.986486	1	0.996621622	1	1	0.993197	0.993007	0.996551	1	1	0.999937	0.999935	0.999968
clusters-4-v1.csv	Logistic Regression	train	0.951389	0.95205479	0.946996466	0.93728223	0.96896552	0.951324752	0.952055	0.940351	0.940559	0.972318	0.951320863	0.952055	0.943662	0.938918	0.970639	0.951318	0.995962	0.9955	0.992256	0.997221	0.995235
clusters-4-v1.csv	Logistic Regression	test	0.954861	0.97101449	0.97222222	0.93333333	0.94444444	0.955253623	0.985294	0.933333	0.945946	0.957746	0.955579969	0.978102	0.952381	0.939597	0.951049	0.955282	0.999599	0.997559	0.99779	0.998377	0.998331
clusters-4-v1.csv	SVC (linear)	train	0.953993	0.9556314	0.950530035	0.94055944	0.96896552	0.953921598	0.958904	0.94386	0.940559	0.972318	0.953910385	0.957265	0.947183	0.940559	0.970639	0.953912	0.995715	0.995504	0.992308	0.99727	0.995199
clusters-4-v1.csv	SVC (linear)	test	0.954861	0.97101449	0.97222222	0.93333333	0.94444444	0.955253623	0.985294	0.933333	0.945946	0.957746	0.955579969	0.978102	0.952381	0.939597	0.951049	0.955282	0.999465	0.997809	0.997474	0.998312	0.998265
clusters-4-v1.csv	SVC (rbf)	train	0.949653	0.95833333	0.944250871	0.92465753	0.97192982	0.949792891	0.945205	0.950877	0.944056	0.958478	0.949654031	0.951724	0.947552	0.934256	0.965157	0.949672	0.996735	0.995528	0.991981	0.997025	0.995317
clusters-4-v1.csv	SVC (rbf)	test	0.958333	0.95714286	0.959459459	0.93506494	0.98507463	0.95918547	0.985294	0.946667	0.972973	0.929577	0.958627806	0.971014	0.95302	0.953642	0.956522	0.95855	0.999532	0.996933	0.997537	0.998248	0.998062
clusters-4-v1.csv	Random Forest Classifier (1)	train	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v1.csv	Random Forest Classifier (1)	test	0.9375	0.91666667	0.957142857	0.93243243	0.94444444	0.9376716	0.970588	0.893333	0.932432	0.957746	0.93852512	0.942857	0.924138	0.932432	0.951049	0.937619	0.99883	0.989546	0.995043	0.997696	0.995279
clusters-4-v1.csv	Random Forest Classifier (3)	train	0.966146	0.97577855	0.958041958	0.95470383	0.97586207	0.966096602	0.965753	0.961404	0.958042	0.979239	0.966109411	0.97074	0.95972	0.95637	0.977547	0.966094	0.999299	0.998786	0.998183	0.999511	0.998945
clusters-4-v1.csv	Random Forest Classifier (3)	test	0.951389	0.95714286	0.971830986	0.93333333	0.94444444	0.951687905	0.985294	0.92	0.945946	0.957746	0.952246636	0.971014	0.945205	0.939597	0.951049	0.951717	0.999265	0.991612	0.997095	0.998475	0.996612
clusters-4-v1.csv	Random Forest Classifier (5)	train	0.959201	0.96885813	0.950704225	0.94482759	0.97231834	0.959177071	0.958904	0.947368	0.958042	0.972318	0.959158207	0.963855	0.949033	0.951389	0.972318	0.959149	0.99877	0.997883	0.997254	0.999162	0.998267
clusters-4-v1.csv	Random Forest Classifier (5)	test	0.954861	0.97101449	0.95890411	0.94666667	0.94366197	0.95506181	0.985294	0.933333	0.959459	0.943662	0.955437221	0.978102	0.945946	0.95302	0.943662	0.955183	0.998797	0.9918	0.996527	0.998248	0.996343
clusters-4-v1.csv	Neural Network Classifier (5)	train	0.930556	0.9556314	0.955719557	0.94423792	0.87460815	0.932549256	0.958904	0.908772	0.888112	0.965398	0.930296463	0.957265	0.931655	0.915315	0.917763	0.9305	0.996034	0.993703	0.990177	0.993304	0.993304
clusters-4-v1.csv	Neural Network Classifier (5)	test	0.940972	0.95714286	0.985714286	0.93150685	0.89333333	0.941924331	0.985294	0.92	0.918919	0.943662	0.941968752	0.971014	0.951724	0.92517	0.917808	0.941429	0.999599	0.995618	0.992991	0.992731	0.995235
clusters-4-v1.csv	Neural Network Classifier (5,5)	train	0.928819	0.94915254	0.93639576	0.94230769	0.89171975	0.929893935	0.958904	0.929825	0.856643	0.968858	0.92855754	0.954003	0.933099	0.897436	0.92869	0.928307	0.9901	0.989846	0.967195	0.984672	0.982953
clusters-4-v1.csv	Neural Network Classifier (5,5)	test	0.927083	0.97101449	0.971830986	0.94117647	0.8375	0.930380487	0.985294	0.92	0.864865	0.943662	0.928455239	0.978102	0.945205	0.901408	0.887417	0.928033	0.999666	0.99543	0.981372	0.977153	0.988405
clusters-4-v1.csv	Neural Network Classifier (5,5,5)	train	0.943576	0.95238095	0.946428571	0.91986063	0.95326646	0.943499153	0.958904	0.929825	0.923077	0.961938	0.943435828	0.955631	0.938053	0.921466	0.958621	0.943443	0.996165	0.994241	0.991134	0.994924	0.994116
clusters-4-v1.csv	Neural Network Classifier (5,5,5)	test	0.954861	0.97101449	0.985714286	0.93421053	0.93150685	0.955611539	0.985294	0.92	0.959459	0.957746	0.955625014	0.978102	0.951724	0.946667	0.944444	0.955234	0.999599	0.996495	0.996969	0.996885	0.997487
clusters-4-v1.csv	Neural Network Classifier (10)	train	0.943576	0.94557823	0.927586207	0.94244604	0.95862069	0.943557793	0.952055	0.94386	0.916084	0.961938	0.943484019	0.948805	0.935652	0.929078	0.960276	0.943453	0.995413	0.995233	0.986781	0.996853	0.99357
clusters-4-v1.csv	Neural Network Classifier (10)	test	0.944444	0.94366197	0.97222222	0.94366197	0.91891892	0.944616271	0.985294	0.933333	0.905405	0.957746	0.945444834	0.964029	0.952381	0.924138	0.937931	0.94462	0.999666	0.995806	0.99659	0.996949	0.997253
clusters-4-v2.csv	Logistic Regression	train	0.863715	0.85964912	0.794326241	0.8419244	0.95578231	0.862920519	0.839041	0.785965	0.856643	0.972318	0.863491926	0.84922	0.790123	0.84922	0.963979	0.863136	0.971392	0.946628	0.960731	0.997137	0.968972
clusters-4-v2.csv	Logistic Regression	test	0.861111	0.80519481	0.881355932	0.81481481	0.95774648	0.864778008	0.911765	0.693333	0.891892	0.957746	0.863684102	0.855172	0.776119	0.851613	0.957746	0.860163	0.979412	0.957371	0.975309	0.999286	0.977845
clusters-4-v2.csv	SVC (linear)	train	0.864583	0.86879433	0.798586572	0.83617747	0.95238095	0.863984831	0.839041	0.792982	0.856643	0.968858	0.86438126	0.853659	0.795775	0.846287	0.960549	0.864067	0.971149	0.946272	0.9608	0.997029	0.968812
clusters-4-v2.csv	SVC (linear)	test	0.864583	0.81333333	0.881355932	0.81927711	0.95774648	0.867928213	0.897059	0.693333	0.918919	0.957746	0.866764389	0.853147	0.776119	0.866242	0.957746	0.863314	0.979078	0.956056	0.976193	0.999156	0.977621
clusters-4-v2.csv	SVC (rbf)	train	0.864583	0.875	0.810035842	0.82119205	0.95189003	0.864529482	0.839041	0.792982	0.867133	0.958478	0.864408482	0.856643	0.801418	0.843537	0.955172	0.864193	0.972416	0.946247	0.960206	0.996488	0.968839
clusters-4-v2.csv	SVC (rbf)	test	0.871528	0.83098592	0.885245902	0.8045977	0.98550725	0.876584191	0.867647	0.72	0.945946	0.957746	0.872834871	0.848921	0.794118	0.869565	0.971429	0.871008	0.978944	0.959687	0.977646	0.998962	0.97881
clusters-4-v2.csv	Random Forest Classifier (1)	train	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
clusters-4-v2.csv	Random Forest Classifier (1)	test																					

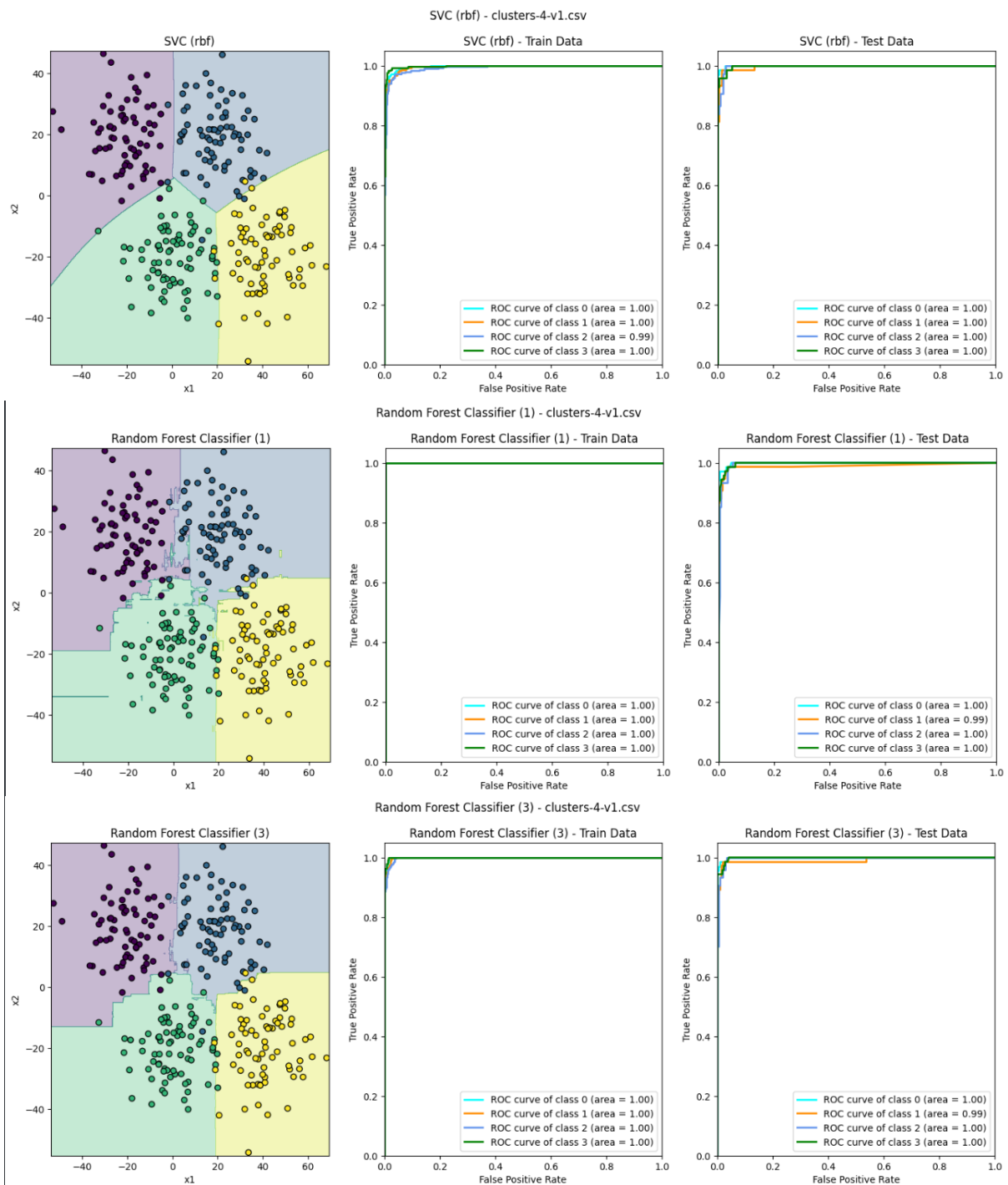
Plots for “clusters-4-v0.csv”

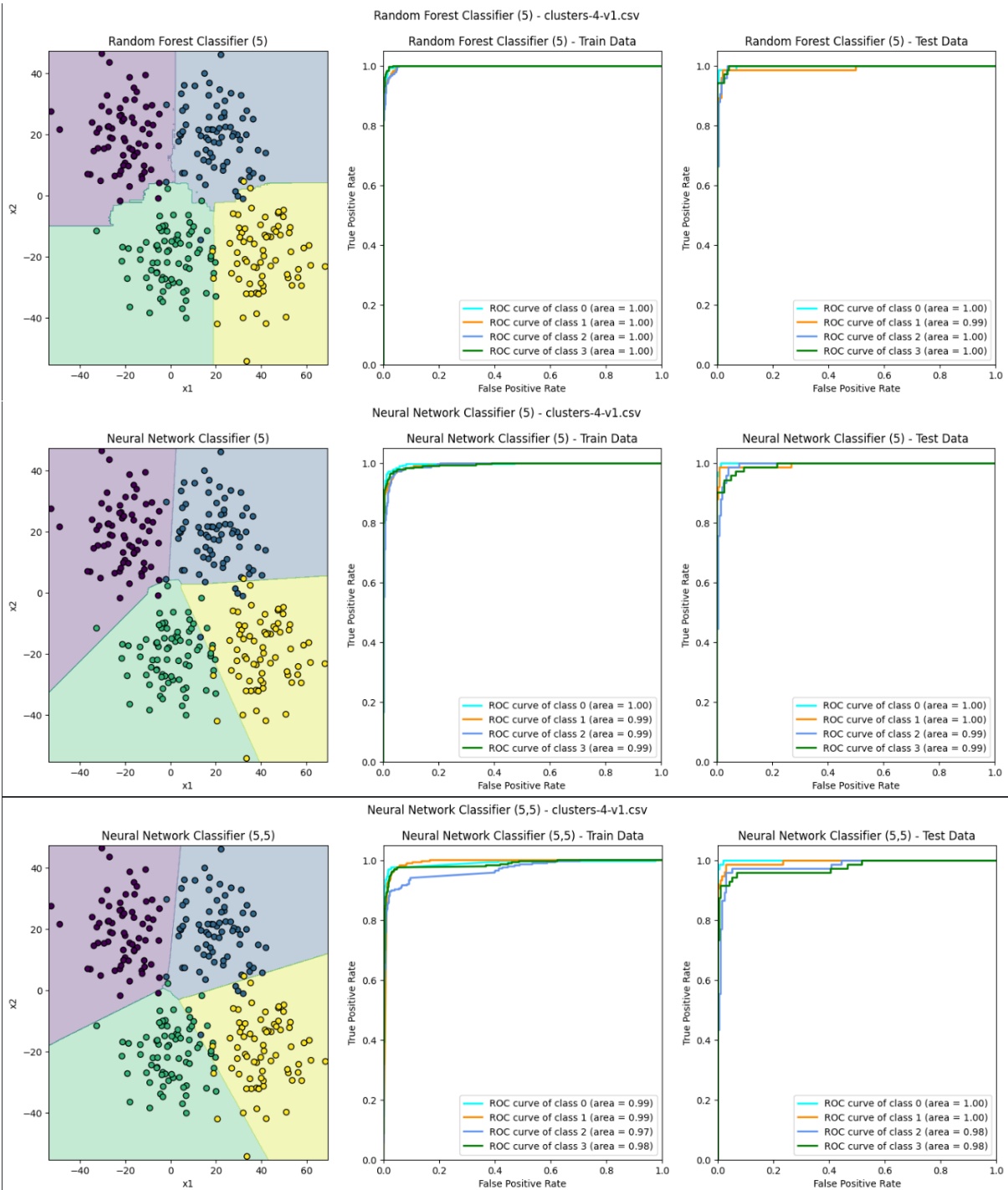


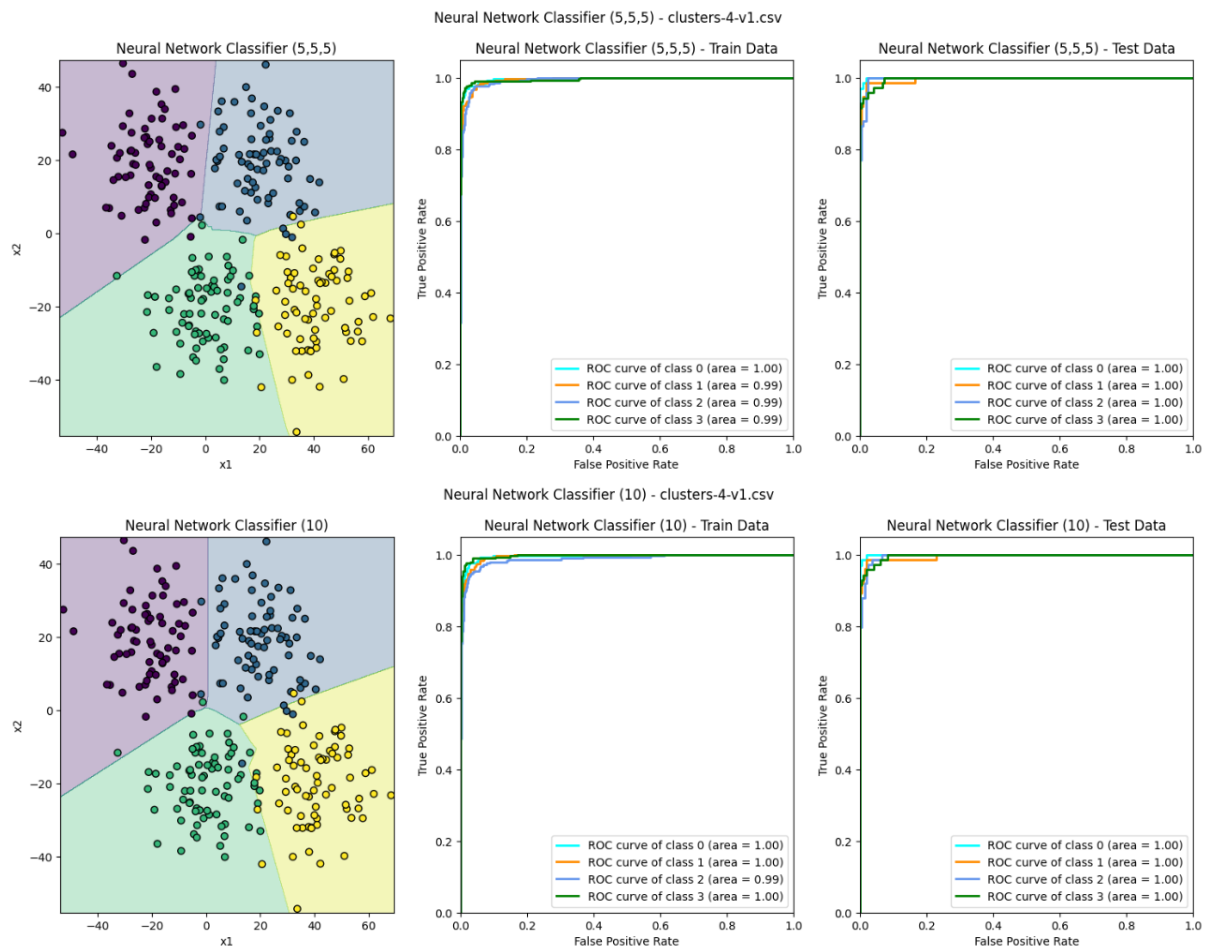




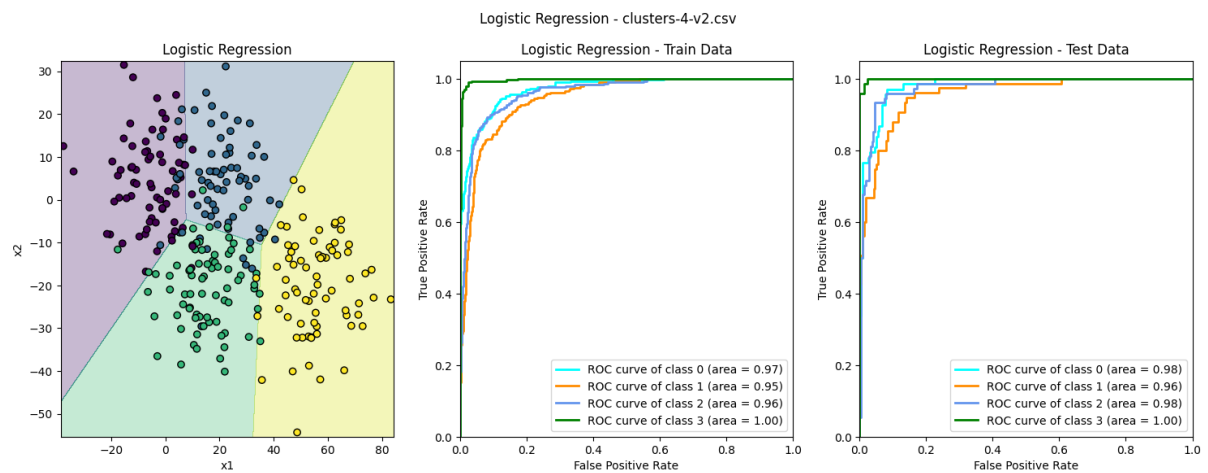


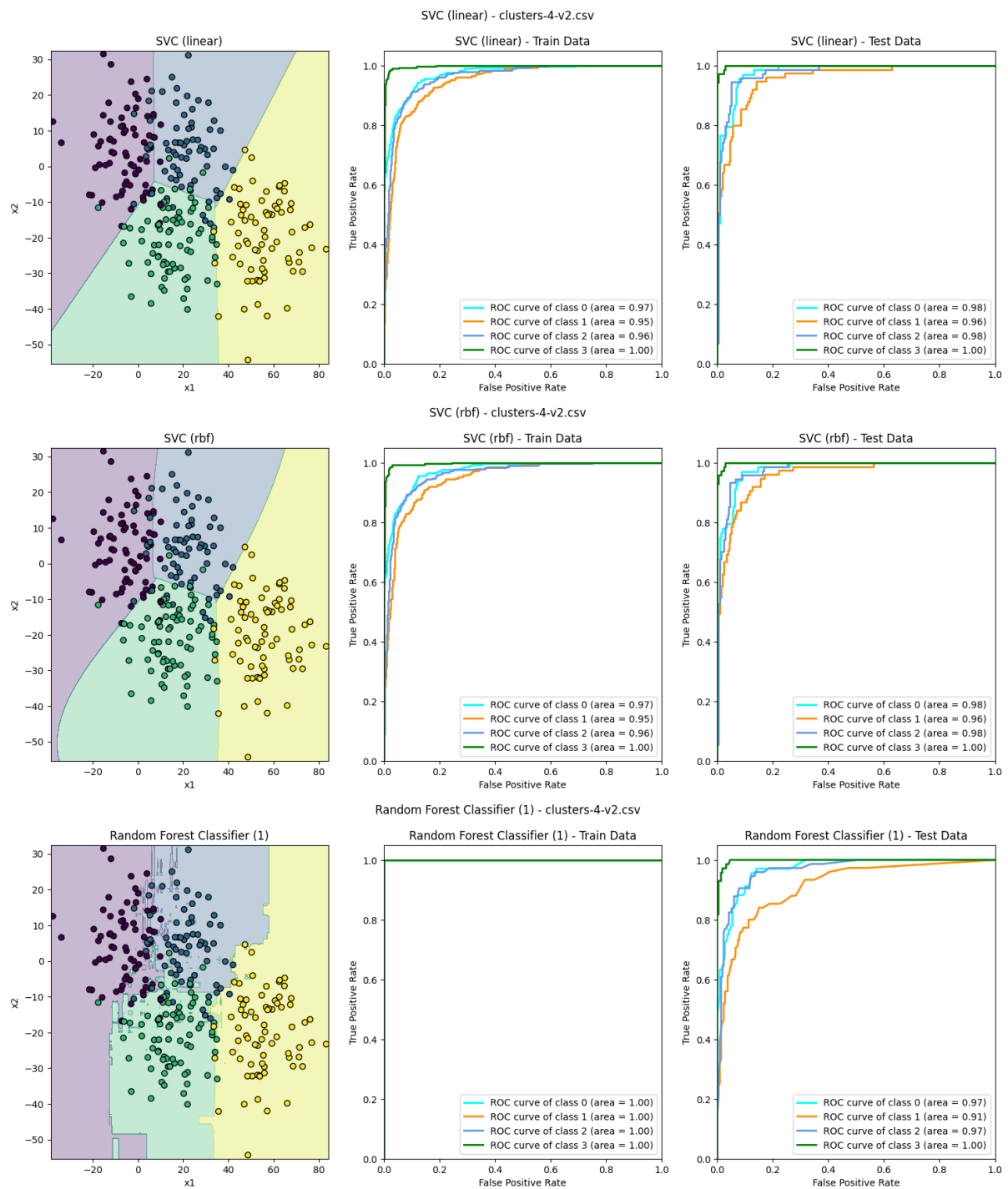


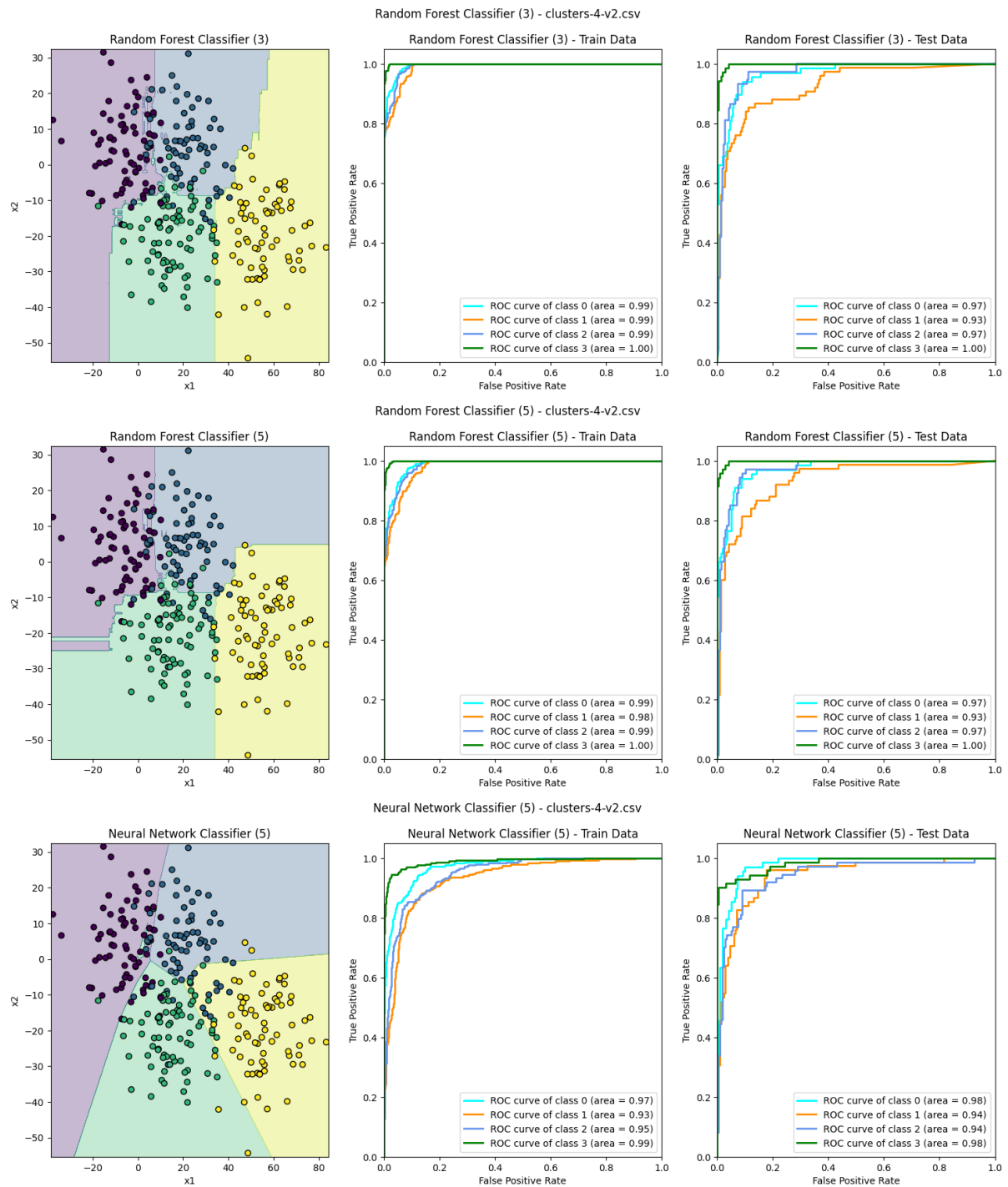


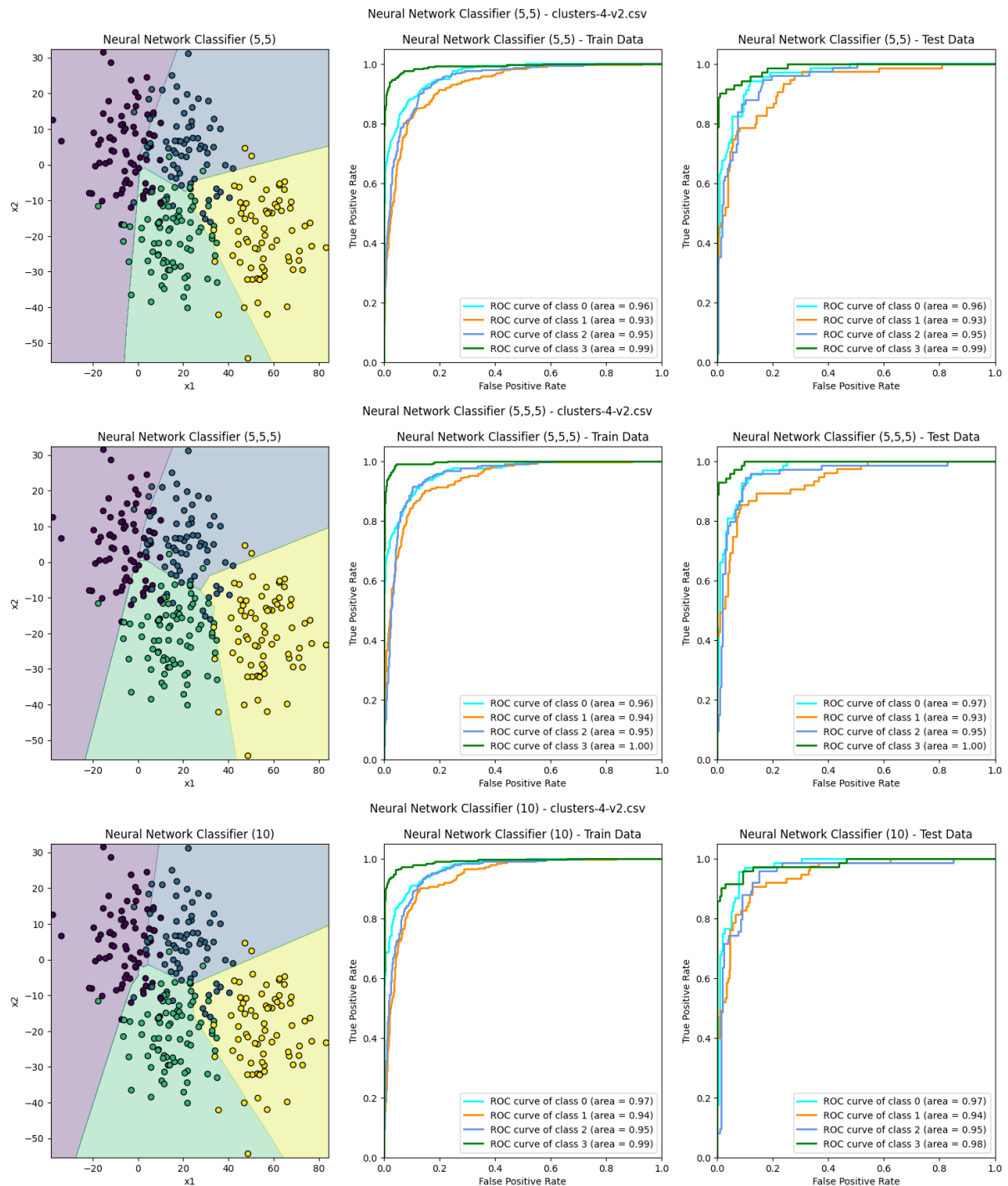


Plots for “clusters-4-v2.csv”







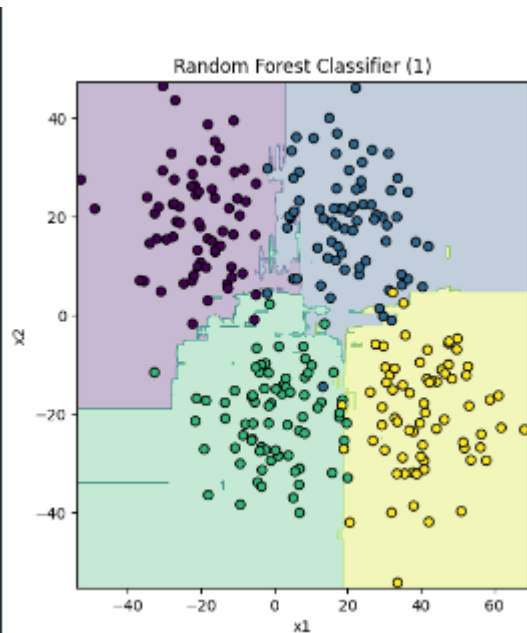


For the clusters-4-v0.csv dataset, the accuracy and precision are 1 for every classification model except Neural Networks (which is also almost 1). This is because the clusters in the dataset are far apart.

Such high accuracy and precision in both the train and test datasets highlight that the model does not overfit the dataset; this can be seen in decision boundaries and the plot.

For the clusters-4-v1.csv dataset, the accuracy and precision are between 0.92 to 0.97, which shows that all the models are still able to identify proper class reasonably well.

Random Forest Classifier with 1 min_samples_leaf overfits the dataset with 1 in every metric for the training dataset and poor performance in the test dataset. This is also clearly visible in the plot



As we can see, the decision boundaries are very complex, showcasing overfitting.

For the clusters-4-v2.csv dataset,

- Logistic Regression has accuracy in the range of 0.86 on both train and test sets.
- SVC models also perform well, with similar results across the linear and rbf kernels, around 0.86-0.87.
- Random Forest Classifier (1) seems to have an overfitting issue, with train accuracy at 1.0, but test accuracy is much lower, around 0.83.
- Neural Networks (with varying layer sizes) display good performance, with most test accuracies in the range of 0.79 to 0.89.

Main Learning:

- Learnt about various Classification Models.
- Used Various Plots to understand Dataset better.